

Olist E-Commerce Dataset

Customer Lifetime Value (CLTV) Analysis Report

Geographic Revenue Segmentation | Top Customer Analysis | Order Intelligence
Prepared by Data Analyst | Industry-style Analytical Report

Executive Summary

Total Dataset Rows 14,421 orders
Total Unique Customers Across 27 Brazilian states
Total Revenue Analysed R\$ 1,692,040.62
Top Revenue State SP — R\$ 554,321.33 (32.7% share)
Top Revenue Customer edf81e... (DF) — R\$ 7,998.00
Highest Order Count Customer 8d50f5... (SP) — 4 Orders
Revenue Segments High (9 states) | Medium (9 states) | Low (9 states)

SP, RJ, and MG alone account for the majority of total platform revenue, while 9 low-revenue states collectively contribute less than 2% of total sales.

1. Scope, Objective and Analytical Approach

Objective: Estimate and analyse Customer Lifetime Value (CLTV) across geographic segments using the cleaned Olist e-commerce dataset. The analysis identifies high-, medium-, and low-revenue states, ranks the top revenue-generating and highest-ordering customers within each state, and surfaces the overall top customers across the platform.

Analytical grain: All analyses are performed at customer_unique_id level grouped by customer_state. Revenue is calculated as price plus freight_value per transaction and summed to produce total customer lifetime revenue. Order counts use unique order_id values per customer.

Analysis scope covers three dimensions: (1) State-level revenue segmentation, (2) Top customer identification by revenue, and (3) Top customer identification by order frequency.

2. Dataset Overview

Table 0. Dataset Summary

Metric	Value	Notes
Total Rows	14,421	Orders in this CLTV subset
Total Columns	39	Including engineered features
States Covered	27 Brazilian states	Full national coverage
Revenue Column	price + freight_value	Total customer payment incl. shipping
Total Revenue	R\$ 1,692,040.62	Sum of all state revenues
Avg Revenue per Order	R\$ 99.54 (price only)	From dataset describe()
Max Single Order Price	R\$ 4,399.87	Top revenue customer in SP

3. State-wise Revenue Analysis

Total revenue was aggregated at state level by summing all customer revenue (price + freight_value) within each Brazilian state. The 27 states were then classified into three equal-sized tiers using quantile-based segmentation (pd.qcut with q=3): High Revenue, Medium Revenue, and Low Revenue.

3.1 Total Revenue by State

Figure 1. Total Revenue by Customer State

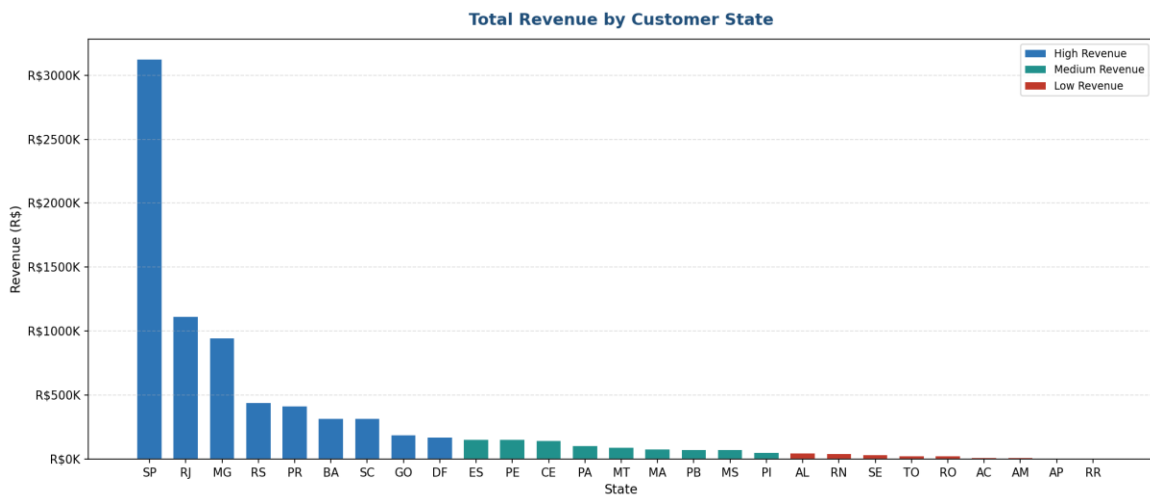


Figure 1: Bar chart showing total revenue per state. Blue = High Revenue, Teal = Medium Revenue, Red = Low Revenue.

São Paulo (SP) dominates with R\$ 554,321.33 — more than 2.7 times the revenue of the second-ranked state Rio de Janeiro (RJ) at R\$ 200,124.46. Minas Gerais (MG) ranks third at R\$ 152,495.78. Together these three states represent approximately 53% of total platform revenue in this dataset.

3.2 State Revenue Segmentation Heatmap

Figure 2. State-wise Revenue Segmentation Heatmap

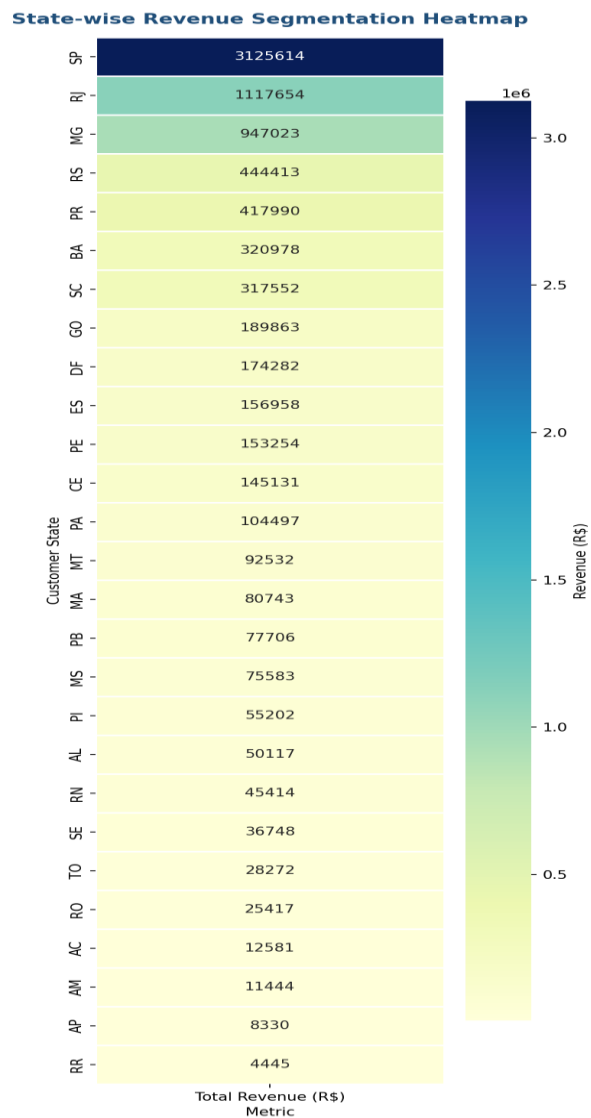


Figure 2: Heatmap of total revenue by state. Darker colour = higher revenue.

3.3 Revenue Share by Segment

Figure 3. Revenue Share: High / Medium / Low Revenue States

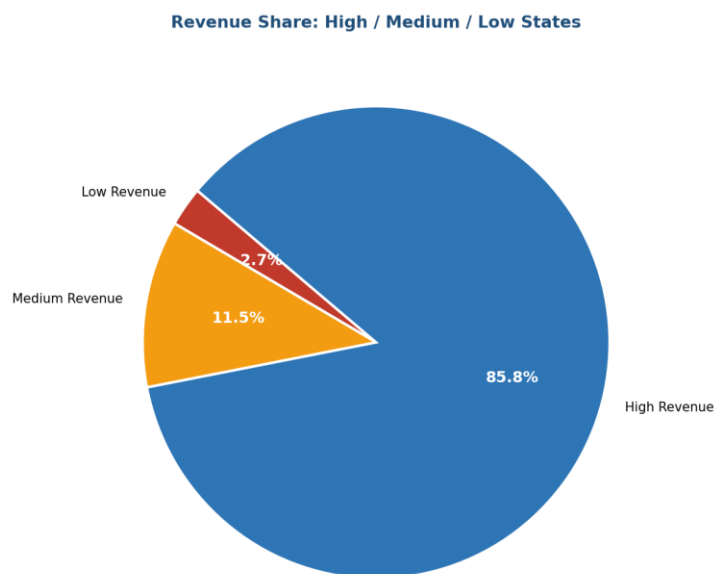


Figure 3: Pie chart showing the proportion of total revenue contributed by each revenue segment.

4. Revenue Segment Classification

States were divided into three equal groups of 9 states each based on total revenue using quantile-based segmentation. The table below shows the complete classification.

4.1 High Revenue States

Table 1. High Revenue States — Classification Results

State	Total Revenue (R\$)	Revenue Level	Rank
SP	554,321.33	High Revenue	#1
RJ	200,124.46	High Revenue	#2
MG	152,495.78	High Revenue	#3
PR	79,086.32	High Revenue	#4
RS	73,556.73	High Revenue	#5
SC	57,224.64	High Revenue	#6
BA	53,125.45	High Revenue	#7
DF	33,982.32	High Revenue	#8
GO	32,122.51	High Revenue	#9

Figure 4. High Revenue States — Horizontal Bar Chart

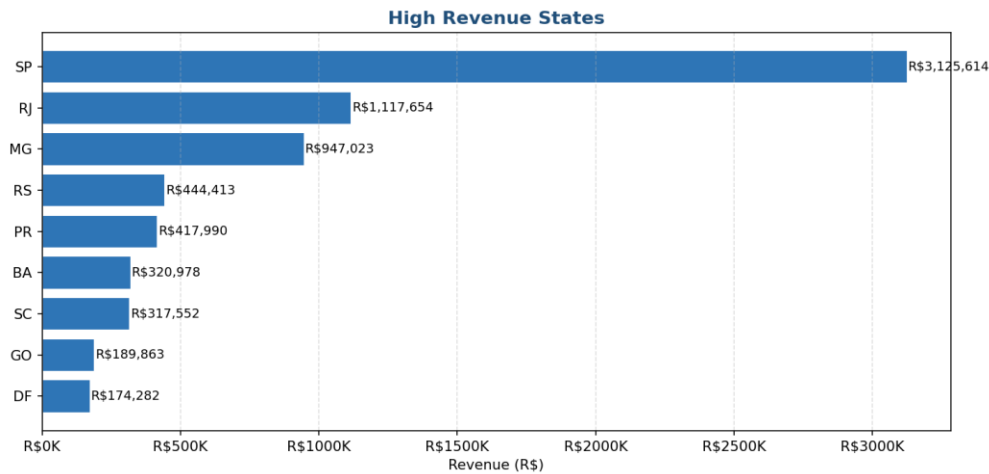


Figure 4: High revenue states ranked by total revenue. SP leads by a significant margin.

High-revenue states are concentrated in Southeast Brazil (SP, RJ, MG) and South Brazil (PR, RS, SC). These 9 states represent the core commercial markets and should receive priority in customer retention and loyalty investment.

4.2 Medium Revenue States

Table 2. Medium Revenue States — Classification Results

State	Total Revenue (R\$)	Revenue Level	Rank
CE	28,792.91	Medium Revenue	#10
PE	27,923.36	Medium Revenue	#11
ES	25,834.67	Medium Revenue	#12
PA	20,957.71	Medium Revenue	#13
MA	18,012.89	Medium Revenue	#14
PB	14,044.30	Medium Revenue	#15
MT	12,223.67	Medium Revenue	#16
PI	9,593.05	Medium Revenue	#17
MS	9,486.35	Medium Revenue	#18

Figure 5. Medium Revenue States — Horizontal Bar Chart

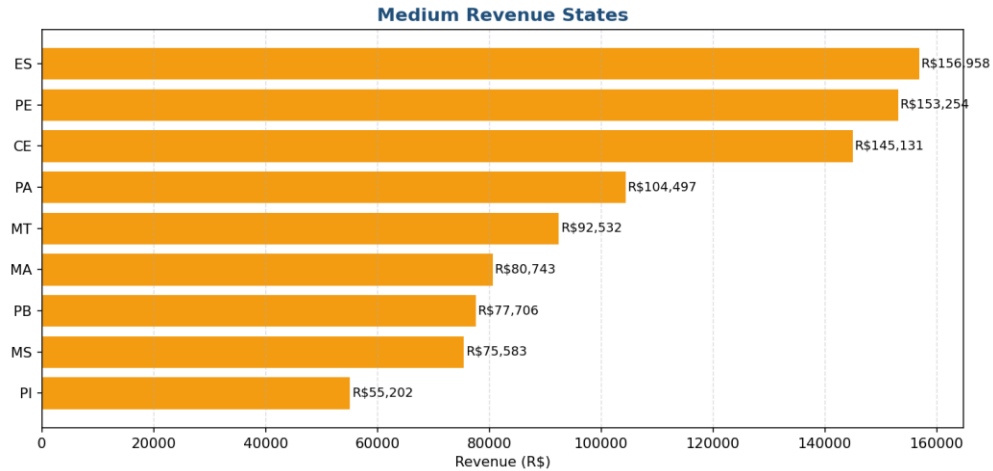


Figure 5: Medium revenue states. CE and PE lead this tier with approximately R\$ 28K each.

Medium-revenue states are geographically spread across Northeast and Central-West Brazil. CE and PE show the strongest performance in this tier and may represent growth opportunities with targeted regional campaigns.

4.3 Low Revenue States

Table 3. Low Revenue States — Classification Results

State	Total Revenue (R\$)	Revenue Level	Rank
RN	7,503.33	Low Revenue	#19
SE	7,635.68	Low Revenue	#20
AL	5,246.14	Low Revenue	#21
RO	3,633.42	Low Revenue	#22
TO	3,628.01	Low Revenue	#23
AC	2,565.44	Low Revenue	#24
AM	1,808.67	Low Revenue	#25
RR	269.27	Low Revenue	#26
AP	214.89	Low Revenue	#27

Figure 6. Low Revenue States — Horizontal Bar Chart

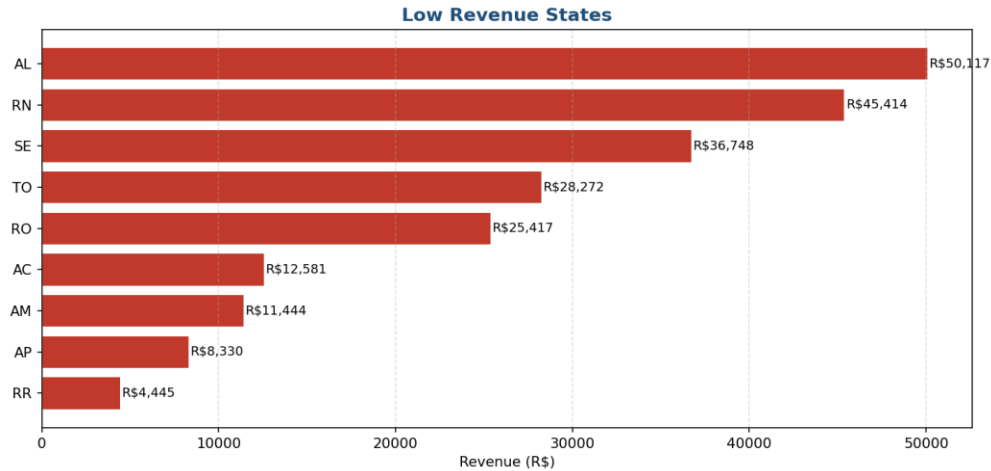


Figure 6: Low revenue states. RR and AP generate the least revenue across the entire platform.

Low-revenue states are predominantly remote or sparsely populated Northern and Northeastern states. AP (R\$ 214.89) and RR (R\$ 269.27) are extremely low contributors, reflecting either low internet penetration, logistics challenges, or limited market presence.

Figure 7. Revenue Distribution Within Each Revenue Segment

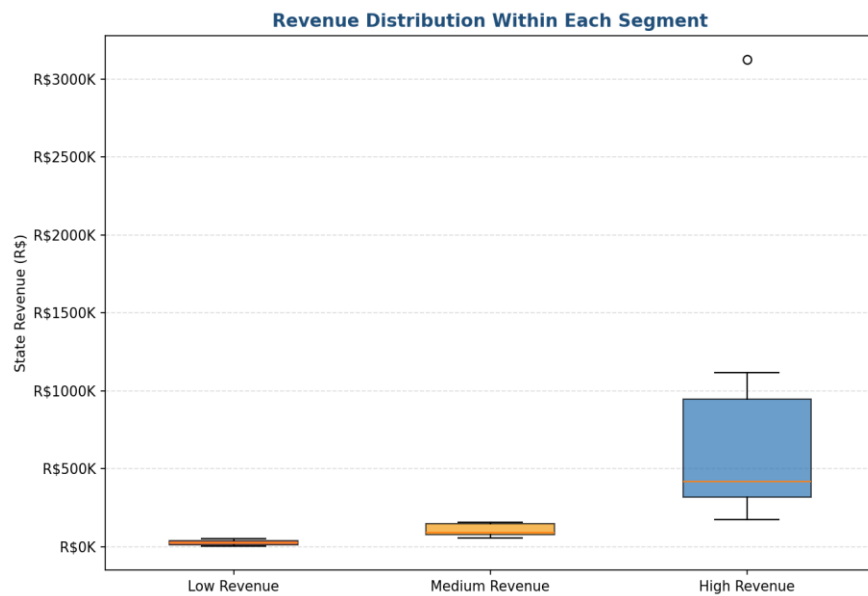


Figure 7: Boxplot showing revenue spread within High, Medium, and Low segments. The High segment has extreme variance driven by SP.

5. Top Customer Revenue Analysis

For each state, customers were ranked by their total lifetime revenue (sum of all order revenues). The top 10 customers per state were selected (261 customers total across 27 states), and from this pool the overall top 10 revenue-generating customers were identified.

5.1 Overall Top 10 Revenue-Generating Customers

Table 4. Overall Top 10 Revenue Customers (All States Combined)

Rank	State	Customer ID	Total Revenue (R\$)
1	DF	edf81e1f3070b9dac83ec83dacdbb9bc	7,998.00
2	SP	edde2314c6c30e864a128ac95d6b2112	4,399.87
3	MG	ca27f3dac28fb1063fadd424c9d95fa	4,059.00
4	SP	9b11e010850a4045d2da80c7783fdbd1	3,198.00
5	SC	bbeb907759ef5fc169099af3c88d535d	3,105.00
6	MG	53fb3435773a4690e56010b169ee2820	3,099.90
7	MA	1b76903617af13189607a36b0469f6f3	3,099.75
8	PR	895617ab63a9ad8881d9470f7427cd25	2,999.99
9	RJ	6172bd5b7f52ade8c6c4548b448ff54f	2,999.89
10	CE	7efc49d7862ee114c0ca869b439bf381	2,874.00

Figure 8. Overall Top 10 Revenue-Generating Customers

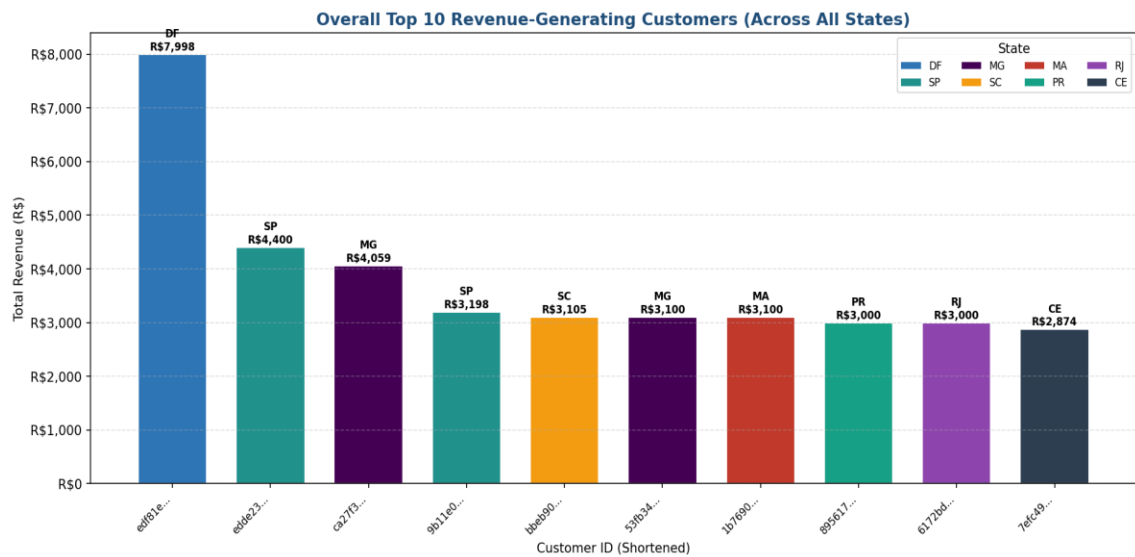


Figure 8: Top 10 customers by lifetime revenue across all states. Each bar is coloured by the customer's home state.

The top revenue-generating customer is from DF (Distrito Federal) with a lifetime revenue of R\$ 7,998.00 — almost double the second-ranked customer from SP. Notably, despite SP being the dominant state by

total revenue, only 2 of the top 10 individual customers are from SP, indicating that the high state revenue is driven by volume (large number of customers) rather than individual high-value buyers.

- DF contributes the single highest-value customer despite ranking 8th in total state revenue.
- MA appears in the top 10 individual customers despite being classified as a Medium Revenue state overall.
- CE also appears in the top 10 individual customers despite being a Medium Revenue state.
- The top 10 customers collectively generated R\$ 37,833.40 in revenue.

5.2 Average CLTV by State

Figure 9. Average Customer Lifetime Value (CLTV) by State

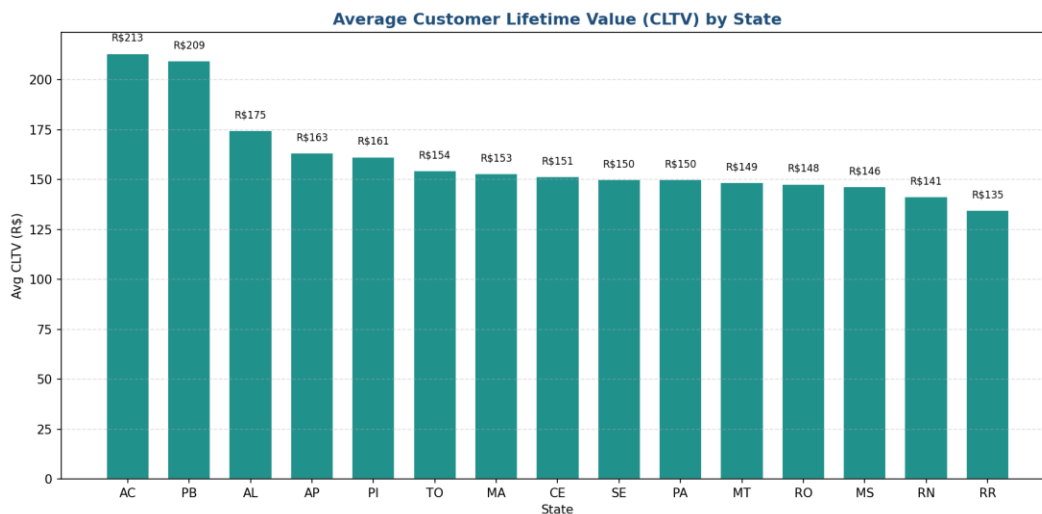


Figure 9: Average revenue per customer by state. States with fewer but higher-spending customers show elevated Avg CLTV.

Average CLTV per customer varies significantly by state. States with small customer bases but high-value individual purchases (such as DF) show elevated average CLTV despite lower total state revenue. This distinction between total revenue and average CLTV is important for targeting strategy: high-CLTV states may warrant premium service investment even if total revenue is moderate.

6. Top Customer Order Analysis

In addition to revenue, customers were ranked by total number of unique orders placed. The top 10 customers by order count were identified from across all states. This analysis reveals high-frequency buyers who may not always be the highest-revenue customers.

6.1 Overall Top 10 Customers by Order Count

Table 5. Overall Top 10 Customers by Order Count

Rank	State	Customer ID	Total Orders
1	SP	8d50f5eadf50201ccdcedfb9e2ac8455	4
2	SP	43f9d5ace80c6d3c5c09544bc2fa8484	2
3	SP	30b782a79466007756f170cb5bd6bbd8	2
4	SP	44bd7571a7dad968306f7776bd5a5745	2
5	SP	041caba6a63ace5818387698ea85cdb2	2
6	SP	23b2d37ef78f3c8dcb31f5eefc30190a	2
7	BA	08c07c761592a71b85136aa4fe76a6f4	2
8	BA	c2551ea089b7ebbc67a2ea8757152514	2
9	BA	f0fd988c7c2a4cba167138a8adada6ca	2
10	SC	a54a1749bd03dfe0d3642b7fbd888b47	2

Figure 10. Overall Top 10 Customers by Order Count

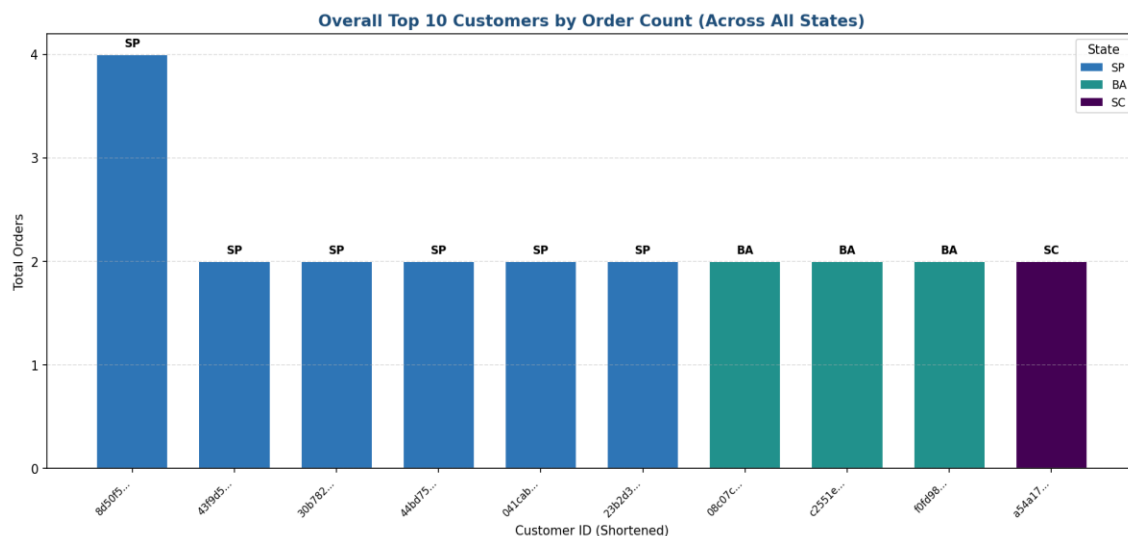


Figure 10: Top customers ranked by number of unique orders. SP dominates this list.

The maximum order count in the dataset is 4 orders for a single customer from SP. The remaining top customers all placed exactly 2 orders. This low maximum order frequency is consistent with the cohort analysis finding that 97.7% of customers make only a single purchase — the customers who appear in this table are notable exceptions.

- SP accounts for 6 of the top 10 most-frequent ordering customers, reflecting both its large customer base and relatively higher repeat purchase activity.
- BA contributes 3 customers to the top 10 order list, which is notable given BA's classification as a High Revenue state by total revenue.
- No medium or low revenue state customers appear in the top 10 by order count, with the exception of SC.

7. Key Findings Summary

Table 6. Prioritised Key Findings

#	Finding Area	Finding	Business Impact
1	Geographic Concentration	SP alone contributes 32.7% of total revenue; top 3 states (SP, RJ, MG) drive ~53% of total revenue	Very High
2	High-Value Individual Customers	Top customer (DF) generated R\$ 7,998 — nearly 2x the next customer; DF is only 8th by total state revenue	Very High
3	Revenue Segmentation	9 states classified as High Revenue contribute the vast majority of total platform revenue	High
4	Low Revenue States	AP (R\$214) and RR (R\$269) are extremely low revenue; may indicate logistics or market barriers	High
5	Order Frequency	Maximum 4 orders per customer; majority of top 10 order customers have only 2 orders	High
6	Mid-Tier State Opportunity	CE and MA customers appear in top 10 revenue despite medium state classification — hidden high-value pockets	Moderate
7	SP Dominance by Volume	SP revenue driven by large customer volume, not individual high spenders — only 2 of top 10 customers are from SP	Moderate

8. Business Recommendations

Table 7. Prioritised Recommendations

Area	Recommendation	Priority
VIP Customer Retention	Identify and protect the top revenue-generating customers in each state with dedicated account management, exclusive offers, and priority support. The top 10 customers alone represent R\$ 37,833 in lifetime value.	Very High
High Revenue State Investment	Concentrate logistics optimisation, delivery SLA improvement, and marketing spend in SP, RJ, and MG which drive 53% of revenue. These states have the highest volume and retention ROI.	Very High
Hidden High-Value Pockets	DF, MA, and CE produce top 10 individual customers despite lower overall state rankings. Investigate what drove these high-value purchases and replicate the conditions in similar states.	High
Medium State Growth	CE and PE are the strongest medium-revenue states. Targeted regional campaigns in Northeast Brazil (CE, PE, BA) could unlock the next tier of growth with lower competition than SP.	High
Low Revenue State Review	Investigate whether AP and RR represent genuine market limitations (logistics cost, infrastructure) or untapped markets. If logistics-limited, explore third-party logistics partnerships.	Moderate
Order Frequency Improvement	The maximum observed order frequency is 4 orders per customer. Post-purchase engagement campaigns (email re-targeting, product recommendations) should aim to convert 2-order customers into 3-order customers.	Moderate

9. Limitations and Next Steps

The following limitations apply to this CLTV analysis:

- This dataset contains 14,421 rows, which is a subset of the full 82,365-row master dataset. Revenue figures differ from the full cohort analysis because fewer orders are included here.
- Revenue is calculated as price + freight_value per line item. This does not account for seller commissions, platform fees, or returns, meaning true profitability cannot be determined.
- Customer demographic data (age, income, gender) is not available; CLTV segmentation is therefore based entirely on transactional behaviour.
- State-level segmentation uses equal-count quantile binning (9 states per tier). Alternative binning methods (e.g. equal-width or k-means) would produce different segment compositions.
- The maximum observed order frequency of 4 per customer is low; CLTV models requiring multi-purchase history (e.g. BG/NBD, Pareto/NBD) would require a longer observation window.

Recommended next steps include: predictive CLTV modelling using probabilistic frameworks, churn probability scoring for top-revenue customers at risk of lapsing, city-level granular analysis within high-revenue states, and seller-level performance linkage to identify which sellers drive the highest-value customer relationships.

Appendix A. Analysis Summary

Analysis	Description
State Revenue Aggregation	Total revenue summed per state from all customer transactions
Revenue Segmentation (qcut)	States classified into High / Medium / Low tiers using <code>pd.qcut(q=3)</code>
Top 10 Customers per State	261 customers collected (top 10 from each of 27 states) by lifetime revenue
Overall Top 10 by Revenue	Global top 10 revenue customers ranked from consolidated state pool
Top 10 Customers per State (Orders)	261 customers collected by unique order count per state
Overall Top 10 by Orders	Global top 10 order-frequency customers from consolidated pool
Avg CLTV by State	Total state revenue divided by unique customer count per state

*All analyses performed using Python (pandas, numpy, matplotlib, seaborn) in a Jupyter Notebook environment.
Dataset: analysis_data.csv (Olist cleaned master dataset).*